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**MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)**
**REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION, APRIL 2026**
**SECOND SEMESTER**
**Multi-Disciplinary Courses (MDC) - MG2MDCMBG100 - BENEFICIAL MICROBES IN DAILY LIFE**
**(2024 ADMISSION ONWARDS)**

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 35**

**Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E),Create(C), Skill(S), Interest(I) and Appreciation(Ap)**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries **1** marks

1. **Explain PGPR?** **[U] / [CO1]**

**Ans:** Definition (1 mark)

2. **Which method involves submerging food in a solution of vinegar, salt, and spices to prevent spoilage?** **[K] / [CO2]**

**Ans:** Pickling

3. **Name one factor affecting human microflora.** **[K] / [CO1]**

**Ans:** Any one factor (1 mark)

4. **Give 2 examples for food preserved by fermentation.** **[U] / [CO2]**

**Ans:** Any two examples (1 mark)

5. **What is the main stimulant found in coffee?** **[K] / [CO2]**

**Ans:** Caffeine (1 mark)

6. **What is the role of microbes in baking?** **[K] / [CO1]**

**Ans:** Fermentation/Flavour and aroma development (1 mark)

7. **What is the role of antioxidants in chemical preservation?** **[K] / [CO2]**

**Ans:** Any one role (1 mark)

8. **How does modified atmosphere packaging help preserve minimally processed food?** **[K] / [CO2]**

**Ans:** By altering composition of gases inside the package, thereby inhibits microbial growth and reduces enzyme activity.

**9. What is the main function of prebiotics? [K] / [CO1]**

**Ans:** It stimulates the growth and activity of beneficial gut microbes (1 mark)

**10. Explain the benefits of using spray drying for food preservation? [A] / [CO2]**

**Ans:** Any two benefits (1mark)

**11. Name the probiotic in yoghurt. [K] / [CO1]**

**Ans:** Lactobacillus

**12. What is the main purpose of nutraceuticals? [K] / [CO2]**

**Ans:** Prevention and management of disease. (1 mark)

**[1x10 = 10]**

## Part B

Short Essay Questions
Answer any **3** questions
Each question carries **5** marks

**13. Explain the factors influencing the survival of microorganisms in air. [U] / [CO1]**

Ans: Any 4 factors such as temperature, relative humidity, UV radiation, Oxygen concentration, Air movement etc (5 marks)

**14. Analyse the health benefit of having wine moderately? [An] / [CO2]**

Ans: Any 4 health benefits in brief (5 marks)

**15. Describe the preparation of curd [K] / [CO2]**

Ans: The steps of Curd production selection of milk → boiling milk → cooling → adding starter culture → incubating → lactic acid fermentation → coagulation**.**

**16. Outline the health benefit of having curd? [U] / [CO1]**

Ans: Any 5 health benefits (5 marks)

**17. Explain the role of microorganisms in cider fermentation. [U] / [CO2]**

Ans: Steps involved and the role of microorganism in each step (5 marks)

**18. What are food preservatives? Differentiate between natural and artificial preservatives with examples. [K] / [CO2]**

Ans: Definition (2 marks)and differentiate ( 3 marks)

**[5x3 = 15]**

## Part C

Long Essay Questions

Answer any **1** question

Each question carries **10** marks

**19. What is the primary purpose of cooking as a method of food preservation, and how does it help to prevent foodborne illness? [K] / [CO2]**

**Ans:** Primary purpose and the ways that helps to prevent foodborne illness in detail (10 marks).

**20. Explain the role of yeast and bacteria in industrial food production. [U] / [CO1]**

**Ans:** Role of yeast in different food industry (5 marks),Role bacteria in different food industry (5 marks)

**[10x1 = 10]**